Medical Image Segmentation Using Minimum Spanning Trees

Minor Assignment Report

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March 11, 2026

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Chapter 1

Introduction

Medical imaging plays an important role in modern healthcare by allowing doctors to examine the internal structures of the human body without invasive procedures. Technologies such as MRI, CT scans, and X-ray imaging help medical professionals detect diseases, analyze tissues, and make accurate diagnoses. However, these images often contain complex details and large amounts of visual information, which makes automatic analysis an important challenge in image processing.

One key step in analyzing medical images is image segmentation, which involves dividing an image into smaller meaningful regions so that structures like organs, bones, or tissues can be identified more clearly. Among various techniques, graph-based segmentation methods have proven to be effective because they model images as graphs where pixels act as nodes connected by edges representing intensity differences. In this assignment, a Minimum Spanning Tree (MST) based segmentation method is implemented to group similar pixels while preserving boundaries between different regions. A segment-based denoising technique is also applied to reduce noise and improve image quality.

Chapter 2

Methodology

2.1 Graph Construction

The first step of the algorithm involves loading and preprocessing the medical image. The image is converted into grayscale so that each pixel is represented by a single intensity value.

Each pixel in the image is treated as a node in a graph. Edges are created between neighboring pixels using a four-neighborhood connectivity model. This means that each pixel is connected to its adjacent right and bottom neighbors.

The weight of each edge is calculated using the absolute difference between the intensity values of two pixels:

$$w(u, v) = |I(u) - I(v)|$$

where $I(u)$ and $I(v)$ represent the intensity values of pixels u and v.

2.2 MST-Based Segmentation

After constructing the graph, all edges are sorted in increasing order based on their weights.

The algorithm uses a Union-Find data structure to maintain connected components efficiently. Initially, each pixel is considered as a separate component.

Edges are processed sequentially. Two components are merged if the edge weight satisfies a threshold condition defined by:

$$w(u, v) \leq \min(Int(C_1) + \frac{k}{|C_1|}, Int(C_2) + \frac{k}{|C_2|})$$

where

- $Int(C)$ represents the internal difference of component C
- $|C|$ represents the size of component C
- k is a constant controlling segmentation sensitivity

Smaller values of k produce many small segments, while larger values produce fewer segments.

2.3 Segment-Based Denoising

After segmentation, a denoising technique is applied. For each segment, the mean intensity value of all pixels belonging to that segment is computed.

Each pixel in the segment is replaced by the segment mean value. This reduces noise while preserving edges between different segments.

Chapter 3

Results and Analysis

The MST-based segmentation algorithm was tested on three medical images:

- Brain MRI image
- Chest X-ray image
- CT scan image

3.1 Evaluation Metrics

The algorithm performance was evaluated using the following metrics:

- Number of segments generated
- Execution time
- Mean Squared Error (MSE)
- Peak Signal-to-Noise Ratio (PSNR)

Image	Segments	Time (s)	MSE	PSNR
Brain MRI	812	0.19	66.84	29.88
Chest X-ray	–	–	–	–
CT Scan	–	–	–	–

Table 3.1: Quantitative performance evaluation

3.2 Segmentation Results

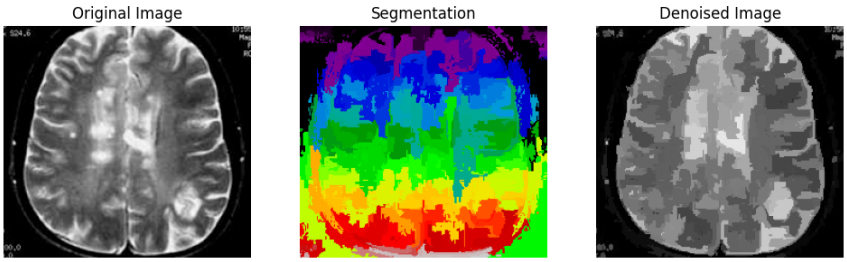

Figure 3.1: Segmentation and denoising results for Brain MRI

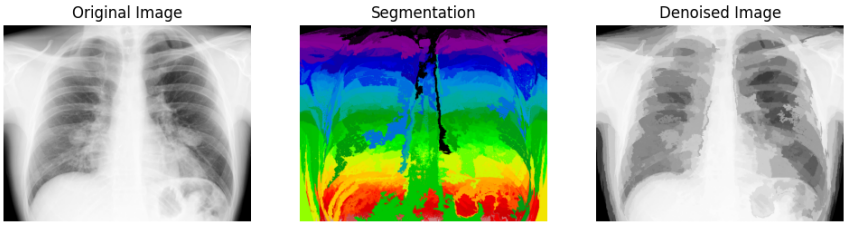

Figure 3.2: Segmentation and denoising results for Chest X-ray

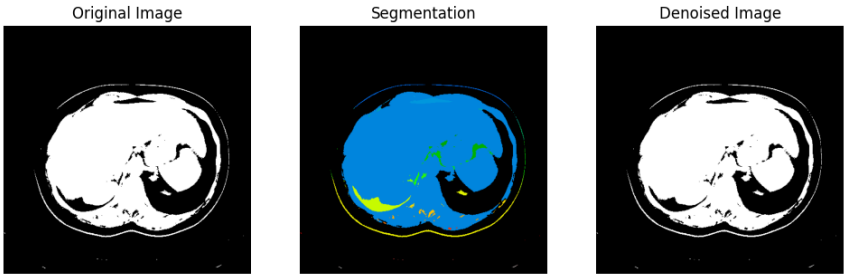

Figure 3.3: Segmentation and denoising results for CT scan

3.3 Segment Distribution Histogram

To better understand how the segmentation algorithm divides the image into regions, histograms were generated showing the distribution of pixels across segments.

Each bar in the histogram represents a segment, and the height indicates the number of pixels belonging to that segment.

3.3.1 Brain MRI Histogram

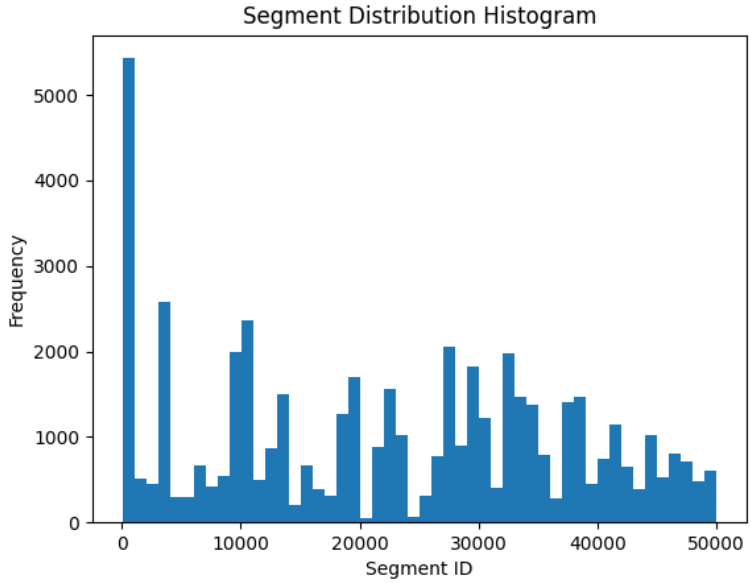

Figure 3.4: Segment distribution histogram for Brain MRI

3.3.2 Chest X-ray Histogram

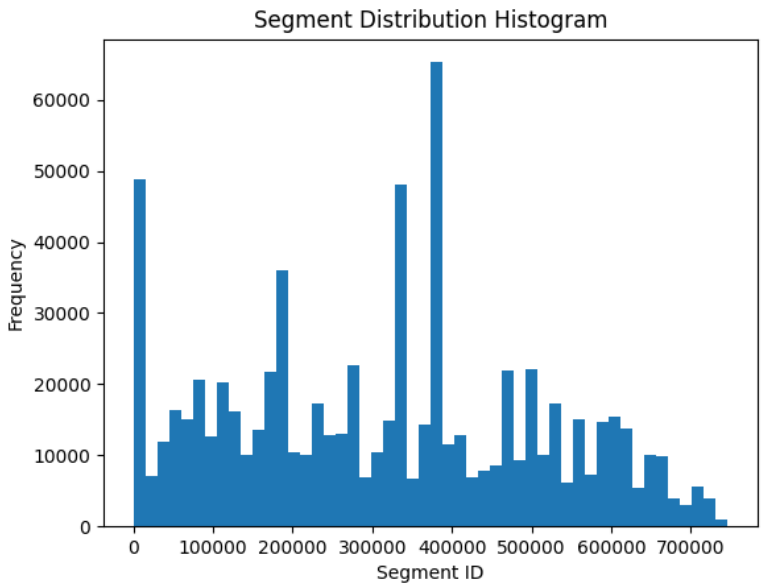

Figure 3.5: Segment distribution histogram for Chest X-ray

3.3.3 CT Scan Histogram

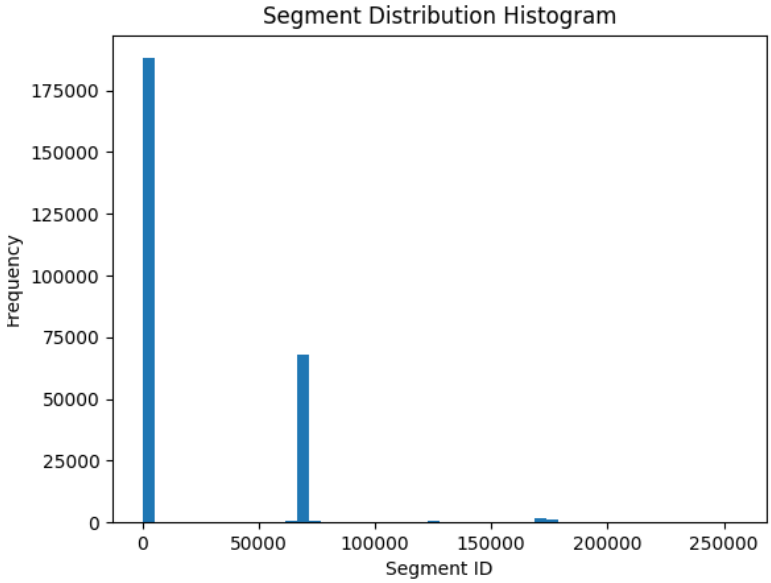

Figure 3.6: Segment distribution histogram for CT Scan

3.4 Parameter Sensitivity

The segmentation parameter k plays an important role in determining segmentation quality. Smaller values of k produce many small segments, while larger values merge more regions together.

Experimental results showed that values between 300 and 500 provide a good balance between detail preservation and noise reduction.

Chapter 4

Discussion

The MST-based segmentation approach provides a simple and efficient way to group pixels with similar intensities while preserving strong image boundaries.

The algorithm performs well for medical images because it adapts naturally to variations in pixel intensity. The Union-Find data structure also ensures efficient computation.

However, the algorithm requires careful tuning of the parameter k. If the value is too small, the image becomes over-segmented. If it is too large, important boundaries may be lost.

Compared with simple thresholding or region growing methods, MST-based segmentation is generally more robust to noise and intensity variations.

Future improvements could involve integrating texture features, multi-scale segmentation, or combining the approach with machine learning techniques.

Chapter 5

Conclusion

This project implemented a graph-based image segmentation technique using Minimum Spanning Trees.

The algorithm successfully segmented medical images by grouping pixels with similar intensity values. Segment-based denoising further improved the image quality by smoothing homogeneous regions while preserving edges.

The results demonstrate that MST-based segmentation is an effective and computationally efficient approach for medical image analysis. With further improvements, the method could be extended to more advanced applications in medical imaging.